higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T100**(E)**(A15)T

NATIONAL CERTIFICATE

BUILDING ADMINISTRATION N5

(4090045)

15 April 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
BUILDING ADMINISTRATION N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

1.1 Name THREE types of power supply used to drive a mechanical plant. (3)

1.2

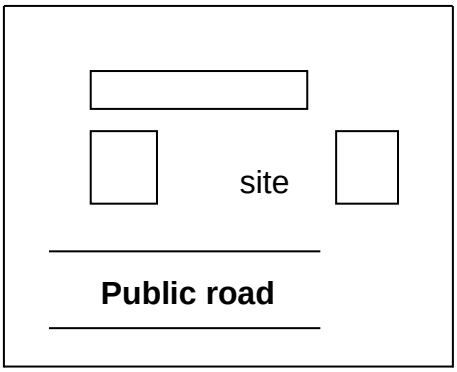


FIGURE 1

FIGURE 1 shows a congested site with 3 properties on it. You were requested to build a structure of 3 floors.

State how you would attend to the following:

1.2.1 Take materials to the third floor

1.2.2 Protect the people walking at the side of the road

1.2.3 The movement of construction vehicles into and out of the site area (access)

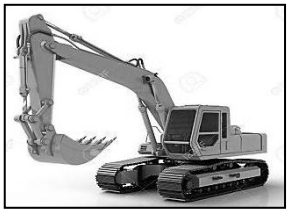
1.2.4 The traffic using the road (public)

1.2.5 Storage of materials

(5 × 1) (5)

1.3 Name the plant or equipment shown BELOW. Write only the answer next to the question number (1.3.1–1.3.10) in the ANSWER BOOK. (10)

1.3.1



1.3.2



1.3.3



1.3.4



1.3.5



1.3.6



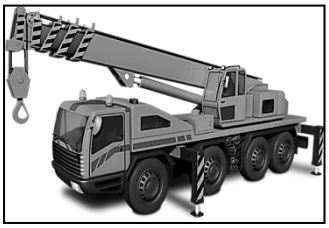
1.3.7



1.3.8



1.3.9



1.3.10



[18]

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QUESTION 2

2.1 State the SIX steps to follow, in sequence, to arrive at effective programming. (6)

2.2 State THREE financial implications you may consider before you take on a project. (3)

2.3 Name the THREE stages of production planning. (3)

2.4 Explain the term *tender appreciation*. Give TWO examples of what should be included in it. (3)

2.5 How is the work progress on a building site measured? (1)

2.6 State FOUR reasons for deviations from the original plan. (4)

2.7 Calculate the number of weeks it will take for a machine with a capacity of excavating 20 m^3/hr to complete a job of 2 850 m^3. Consider an 8-hour day and a 6-day week. (2)

[22]

QUESTION 3

3.1 State FOUR duties of a main contractor regarding the subcontractor. (4)

3.2 Why is piling always mentioned as a separate clause in a contract? (1)

[5]

QUESTION 4

4.1 State THREE fundamental rules that you would follow for the smooth and effective running of a project, when you are in charge of the buying department in a firm and you have to order and purchase materials. (3)

4.2 What should you do when you receive material delivered on site and some of the materials are damaged? (1)

4.3 State FIVE methods that can be implemented to control loss, theft and damage of materials on site. (5)

[9]

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QUESTION 5

5.1 State the procedure and SIX main items usually discussed at a site meeting. (6)

5.2 Name FIVE qualities of a good site foreman. (5)

5.3 State THREE needs that can be identified in human behaviour. (3)

5.4 Define the following terms:

5.4.1 Practical completion

5.4.2 Final completion

5.4.3 Provisional sums

5.4.4 Variation order

5.4.5 Patent defects liability period

5.4.6 Arbitration

(6 × 2) (12)
[26]

QUESTION 6

6.1 Name THREE factors that you should consider when you delegate work to a person. (3)

6.2 What factors would indicate efficient site administration? (3)

6.3 Write the difference between *period planning* and *stage planning*. (2)
[8]

QUESTION 7

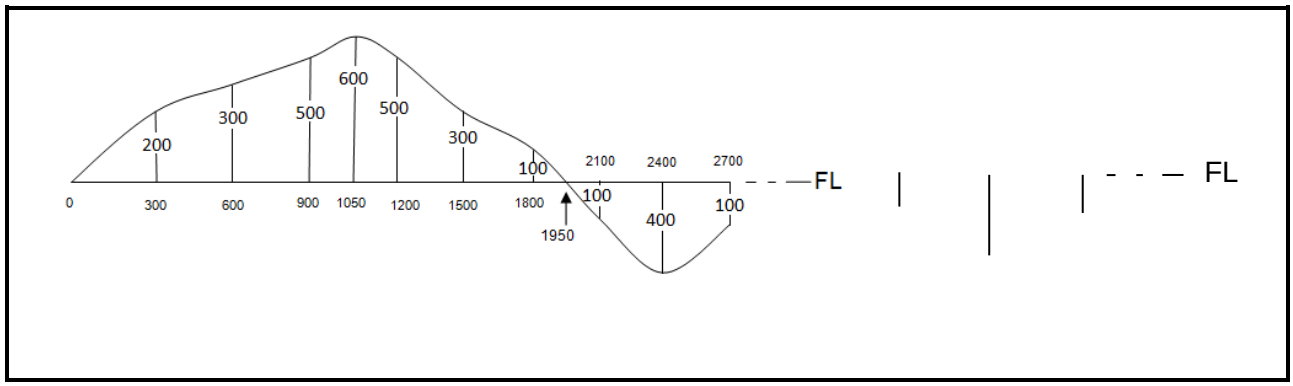


FIGURE 2

The profile in FIGURE 2 shows the cross section of a road that is 2,7 m wide. The length of the road is 15 m.

Use the midordinate method and calculate the volume of earth to be removed after the fill is completed. All values should be approximated to 2 decimal places. **[12]**

TOTAL: 100

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